Customer Behaviour Analysis

Analysing 3,900 transactions to unlock insights on spending patterns, preferences, and subscription behaviour

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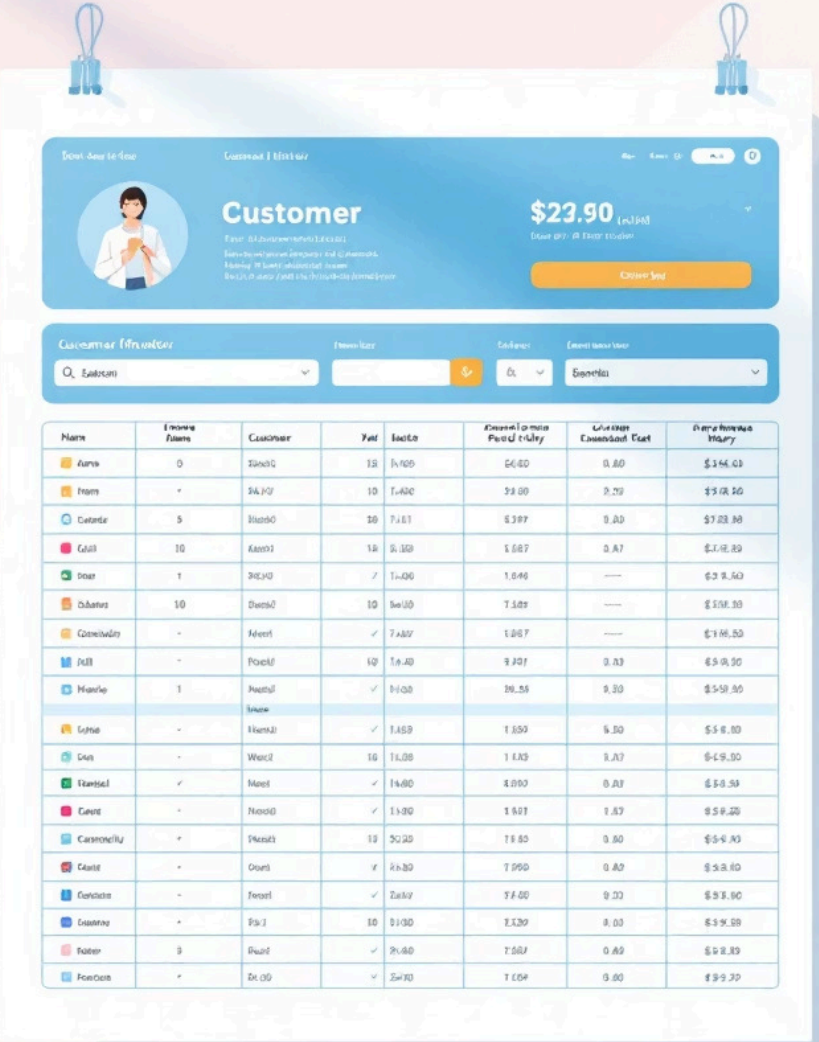
Dataset Overview

3,900

Transaction Records

18

Data Attributes



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Data Structure

Customer Demographics

- Age, gender, location
- Subscription status

Purchase Information

- Product category
- Purchase amount, season

Shopping Behaviour

- Discount applied
- Review ratings

Review Rating column: 37 missing values imputed using median by product category

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Data Preparation Process

Data Import

Loaded dataset using Pandas in Jupyter Notebook

Structure Check

Used df.info() and describe() for initial analysis

Missing Data

Imputed Review Rating using median by category

Standardization

Rename columns to snake case for consistency

Integration

Loaded cleaned DataFrame into PostgreSQL for analysis

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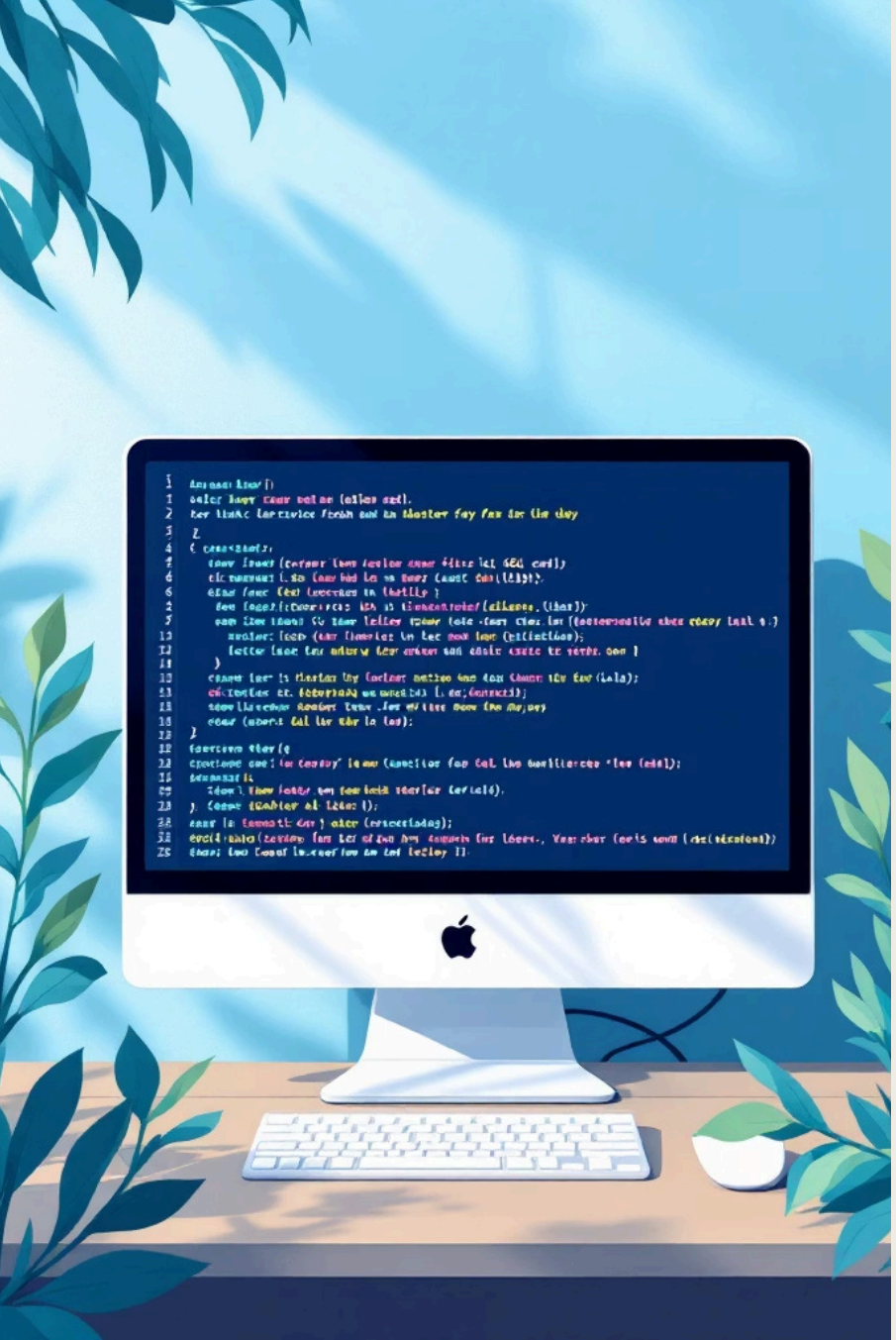

SQL Analysis Approach

1 Top 5 Highest-Rated Products

Average ratings by product category

2 Revenue by Gender

Total revenue comparison: male vs. female customers

3 High-Spending Discount Users

Identified customers using discounts but spending above average

4 Shipping Type Comparison

Average purchase amounts: Standard vs. Express

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Key Analysis Findings

Subscribers vs Non-Subscribers

Compared average spend and total revenue across subscription status

Customer Segmentation

Classified as New, Returning, or Loyal based on purchase history

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Additional Insights

Discount-Dependent Products

Top 5 products with highest percentage of discounted purchases

Repeat Buyers & Subscriptions

Customers with >5 purchases more likely to subscribe

Top Products per Category

Most purchased items within each product category

Revenue by Age Group

Total revenue contribution by age demographics

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Power BI Dashboard

01

Connected to SSMS database

02

Built interactive visualizations

03

Created customer segments

04

Published business intelligence

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Actionable Insights

Increase Subscription Adoption

Offer exclusive deals, early access, and loyalty benefits to drive subscriptions

Strengthen Customer Retention

Implement loyalty programmes and rewards to convert returning customers into loyal buyers

Optimise Discount Strategy

Analyse discount usage to drive sales whilst maintaining healthy profit margins

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Strategic Recommendations

Promote High-Performing Products

Focus marketing campaigns on top-rated and best-selling items

Data-Driven Marketing

Target high-spending age groups and customers preferring express shipping

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